MealSwipe: Health-Aware Meal Planning

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Problem

Deciding what to cook every week is tedious, so people repeat the same few dishes or buy on impulse. Recipe platforms only help if you already know what you want and ignore budget, time, or allergies. Health ratings like the Nutri-Score are one-size-fits-all: a 900 kcal high-protein meal suits someone building muscle but not someone trying to lose weight.

Idea

A web app that finds meals by swiping: a few coarse questions ("Noodles today?"), then dish cards swiped left or right. From the liked dishes it builds a plan of X meals, weighing price, cooking time, and fit with personal goals (e.g. high protein), plus a priced shopping list.

- **Web Client** (Flask/Jinja, plain JS for swiping): swipe UI, profile, plan, list.
- **Preference Engine**: holds the swipe session and picks the next question or card.
- **Recipe Catalog**: recipes, nutrition, and prices, filled via import adapters (mock data first).
- **User Profile**: goals, hard filters (vegan, allergies), and history for variety.
- **Health Evaluator**: rates dishes *relative to the user's goals*; a rule-based and an AI (LLM) implementation with personalized explanations share one interface.
- **Meal Planner**: ranks candidates, selects X meals, aggregates the shopping list.

Style: modular monolith with ports and adapters (replaceable data sources and AI provider). **Quality attributes:** *modifiability* (swap evaluators or sources), *availability* (rule-based fallback if the AI fails), *privacy* (only anonymized nutrient data reaches the LLM), *responsiveness* (next card < 200 ms). **Out of scope:** live prices, ordering, medical diets, native apps.

Benefits

- **Reduced planning effort** – addresses tedious meal selection; shown by the time to create a weekly meal plan and shopping list vs. manual planning (user study).
- **Budget control** – addresses impulse buys; costs are known before shopping, shown by the share of generated plans within the user's budget.
- **Healthier, goal-fitting diet** – addresses one-size-fits-all ratings; shown by one dish rated differently per goal profile and plans meeting calorie/protein targets within 10 %.
- **More variety and satisfaction** – addresses repetitive meals; shown by distinct dishes per user across sessions and participants' ratings of their generated plans.

Action

1. *Wk 1–3, Design:* user stories, quality scenarios, interfaces, ADRs; mock data (100 recipes).
2. *Wk 4–7, Core:* catalog, profile, swipe UI, planner, rule-based evaluator.
   Minimum deliverable: swipe to X meals with priced shopping list and health rating.
3. *Wk 8–10, Extend:* LLM evaluator, coarse questions, recommendations, gamification.
4. *Wk 11–12, Evaluate:* user study, quality scenario checks; buffer.

Tech: Python/Flask, PostgreSQL, Docker, LLM API. **Team:** 2 frontend, 2 backend, 2 health/AI.

AI usage statement. Claude (Anthropic) was used to restructure and rephrase our draft.